HOPE V2 / minimal upstream review

V2 minimal-branch engineering report

This report replaces source-scaffold receipts as a description of the new minimal branch. Historical reports/PDFs for 9118a491 or 2bb6cc56 are not current implementation evidence. The uploaded implementation specification supplies the acceptance contract, not passing results.

Architecture, physical assumptions, compatibility policy, complete file inventory and test evidence are linked from the adjacent implementation, physics, scope, parity and acceptance documents. All 50 A/B/C/T/X identifiers remain visible. Mixed requirements retain separate forecast/software, downstream and physical-data dispositions.

Scope decisions: isolate FlightSpinEstimator; retain the small physical history/config-hash helper; keep the original production estimator/predictor/receiver/packetizer/Python code; use the original one-shot trigger; mark unsupported historical events rather than substituting V2 output; reject unmatched moving-plane datasets. No physical coefficients were tuned.

Report refresh checklist:

- [x] Refresh upstream/source pins and inspect both repository histories.
- [x] Preserve source branches and original production bytes.
- [x] Selectively port physical contracts; exclude PR49 command/selector graph.
- [x] Attach final acceptance/config/code identities and exact current test receipts.
- [x] Report selected 150/300 Hz and optional-profile load results individually.
- [x] Generate an updated PDF from this Markdown and the current acceptance summary.
- [ ] Collect independent real-flight data with clock/frame/registered-plane receipts before physical qualification.

Do not forward earlier PDFs as if they describe this new branch. No original review PDF was supplied in this task, so exact PDF page citations are unavailable; the complete appendix requirement text was supplied in the implementation plan.

Current validation receipt

Tested code: `dcd17786f5b32afa56da63713c5cd32c3dffdd78`. Upstream base: `8d599e7d21e196275cf426488555e2648e8fd789`. Source: `9118a491df658fa7e68f042581af48890ec8b0b8`. Later commits update review documents/evidence only.

All **50 forecast/physics software cases** A01–A14, B01–B12, C01–C08, T01–T06 and X01–X10 PASS. The JSON names each executed group/assertion, configuration hash, command and evidence path. Seventeen downstream clauses remain explicitly deferred; the independent physical-data clause is NOT_RUN.

Clean HOPE-only Ubuntu 24.04/Jazzy build passed with no private checkout mounted. CTest groups: hope_msgs **2/2**; hope_planner_cpp **30/30** (including 35 V2 core assertions, 8 transport assertions and independent installed API consumer); hope_bringup **24/26**, with only inherited flake8/pep257 failures. The original baseline has planner 17/17 and bringup 23/25. Both branches have the same 3,086 flake8 and 37 pep257 file/code issues. One underlying test skips the standalone adapter's unsupported replay-launch mode on both branches; it is not counted as new coverage. ASan/UBSan **13/13** native groups passed.

Strict independently compiled production parity: 18 racket inversion fixtures, 3 receiving-flight cases, 540 predictor cases across all five legacy modes, and 140 spin/batch samples have exactly zero difference. Nakashima and venue contact modes remain distinct. Existing production sources/defaults/command types are pinned; BallSample grows and dependent C++ packages must be rebuilt.

Numerical gate: 42 DOP853 contact comparisons pass the 1 mm / 0.5 ms budgets, plus native analytic plane/apex and geometry/covariance checks. Matched synthetic tooling produced 270 rows, one actual one-shot epoch and six visibly synthetic plots; historical unsupported event paths are not fabricated.

Selected full/packet cadence: **50 Hz**. Compact physical-forecast coverage must remain >=95%; every published compact forecast source age <=30 ms and receipt P95 <=30 ms. Repeated isolated measurements:

Nominal input Hz	Actual observed Hz	Compact coverage	Receipt P95 ms	Full received Hz
150	143.42	100.00%	1.97	49.95
300	288.55	99.14%	1.69	48.75

The clean CTest run independently passed coverage 99.70%/97.41% at nominal 150/300 Hz. A 12 ms delayed worker every 100 solves also passed (99.40%/96.98%). Both directions and both plane policies passed at 150/300 Hz, including automatic serve acquisition, two contacts, spin dropout, context/clock/frame resets and exact source stamps. Full payload/coverage, latency quantiles, CPU/RSS, superseded work and queue bounds are reported separately in the JSON.

Optional unrestricted full output at nominal 300 Hz **fails** compact coverage (91.52%). Optional 100/150 Hz profiles passed this run; they do not replace the conservative default. Earlier broad tests with concurrent builds failed coverage/target timing and remain archived as contention diagnostics; isolated final validation did not waive any thresholds.

Physical accuracy = NOT_RUN. No independent held-out annotated ball flights with clock/frame/registered-plane receipts were supplied. No live calibration, physical superiority, deployment readiness or GitHub Actions result is claimed. Legacy replay comparison is explicitly limited to fixed x_hit=.15 sessions.

Raw current-instance artifacts: `/workspace/.cloud-setup/v2-evidence`. Committed summaries are under `docs/flight_prediction_evidence/`; exact commands are in `flight_prediction_validation.md`.

Software case traceability

The complete assertion names, split clauses, config/code hashes and commands are in the JSON. These are executed software results; physical qualification is separate.

Case	Status	Executed groups
A01	PASS	test_flight_prediction_v2
A02	PASS	test_flight_prediction_v2
A03	PASS	test_flight_prediction_v2
A04	PASS	test_flight_prediction_v2
A05	PASS	test_flight_prediction_v2
A06	PASS	test_flight_prediction_v2
A07	PASS	test_flight_prediction_v2
A08	PASS	test_flight_prediction_v2
A09	PASS	test_flight_prediction_v2, flight_v2_ros_transport, flight_v2_raw_scenarios, test_adaptive_flight_commit, test_incoming_trajectory, test_planner_node_turn_guard
A10	PASS	test_flight_prediction_v2
A11	PASS	test_flight_prediction_v2, flight_v2_raw_scenarios
A12	PASS	test_flight_prediction_v2, test_flight_transport_v2, flight_v2_ros_transport
A13	PASS	flight_v2_ros_transport, flight_v2_causal_replay, flight_v2_raw_scenarios
A14	PASS	test_flight_prediction_v2
B01	PASS	test_flight_prediction_v2, flight_v2_raw_scenarios
B02	PASS	test_flight_prediction_v2
B03	PASS	test_flight_prediction_v2, flight_v2_baseline_epochs
B04	PASS	test_flight_prediction_v2, test_flight_transport_v2
B05	PASS	test_flight_prediction_v2, test_flight_transport_v2, planner_full_rate_relay_test, flight_v2_raw_scenarios
B06	PASS	test_flight_prediction_v2
B07	PASS	test_flight_prediction_v2, test_flight_transport_v2
B08	PASS	test_flight_prediction_v2
B09	PASS	test_flight_prediction_v2
B10	PASS	test_flight_prediction_v2, flight_v2_numerical, flight_v2_baseline_epochs
B11	PASS	flight_v2_causal_replay, flight_v2_scoring, flight_v2_accuracy_tooling, flight_v2_matched_replay_tooling
B12	PASS	test_flight_transport_v2, test_prediction_event_order
C01	PASS	test_flight_prediction_v2

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| C02 | PASS | test_flight_prediction_v2 |
| C03 | PASS | test_flight_prediction_v2 |
| C04 | PASS | test_flight_prediction_v2, test_flight_transport_v2 |
| C05 | PASS | test_flight_prediction_v2 |
| C06 | PASS | test_flight_prediction_v2 |
| C07 | PASS | test_flight_prediction_v2 |
| C08 | PASS | test_flight_prediction_v2 |
| T01 | PASS | test_flight_prediction_v2, test_flight_transport_v2, flight_v2_causal_replay |
| T02 | PASS | test_flight_prediction_v2 |
| T03 | PASS | test_flight_prediction_v2, test_flight_transport_v2, flight_v2_ros_transport |
| T04 | PASS | test_flight_prediction_v2 |
| T05 | PASS | test_flight_prediction_v2 |
| T06 | PASS | test_flight_prediction_v2, test_flight_transport_v2 |
| X01 | PASS | test_flight_prediction_v2 |
| X02 | PASS | test_flight_prediction_v2 |
| X03 | PASS | test_flight_prediction_v2 |
| X04 | PASS | test_flight_prediction_v2, flight_v2_ros_transport, flight_v2_raw_scenarios |
| X05 | PASS | flight_v2_raw_scenarios, flight_v2_scope |
| X06 | PASS | test_flight_transport_v2 |
| X07 | PASS | test_flight_transport_v2, flight_v2_ros_transport |
| X08 | PASS | test_flight_prediction_v2, flight_v2_scope, flight_v2_baseline_epochs,
flight_v2_binary_scope, flight_v2_installed_api |
| X09 | PASS | test_flight_prediction_v2, test_flight_transport_v2, flight_v2_numerical |
| X10 | PASS | flight_v2_scope, flight_v2_binary_scope, flight_v2_ros_transport |
```